

DATA CLEANING

TIPS FOR ANALYSIS



Tip 1

Handle Missing Data

Missing values can skew your analysis.

- Fill missing data with mean, median, or mode, or remove them if appropriate.

```
data.fillna(data.mean(), inplace=True)  
data.dropna(inplace=True)
```

Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	NaN	2022-08-15
Sam Brown	NaN	45000	NaN
Lisa Ray	27	60000	2023-02-20



Name	Age	Salary	Join Date
John Doe	29.00	50000.00	2023-05-10
Jane Smith	34.00	51666.67	2022-08-15
Lisa Ray	27.00	60000.00	2023-02-20

Tip 2

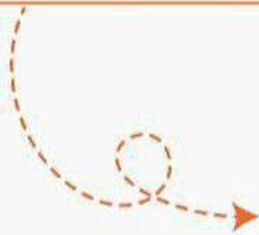
Remove Duplicates

Duplicates can distort your results.

- Use `.drop_duplicates()` to clean up.

```
data.drop_duplicates(inplace=True)
```

Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2023-02-20
John Doe	29	50000	2023-05-10



Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2023-02-20

Tip 3

Fix Data Types

Incorrect data types can lead to errors.

- Convert columns to the appropriate type, like strings to dates or integers.

```
data['date'] = pd.to_datetime(data['date'])
```

Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	15-Aug-2022



Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2022-08-15

Tip 4

Standardize Formats

Inconsistent formats make merging datasets harder.

- Use `.lower()` for text or `.strftime()` for dates.

```
data['name'] = data['name'].str.lower()
```

Name	Age	Salary	Join Date
JOHN DOE	29	50000	2023-05-10
jane smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2022-08-15



Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2022-08-15

Tip 5

Remove Outliers

Outliers can distort analysis.

- Use statistical methods like IQR to detect and remove outliers.

Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2022-08-15
Sam Brown	28	1000000	2023-03-01



Name	Age	Salary	Join Date
John Doe	29	50000	2023-05-10
Jane Smith	34	51666.67	2022-08-15
Lisa Ray	27	60000	2022-08-15